

Ryder DeBack

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EDUCATION

University of Washington Tacoma | *School of Engineering and Technology*
Bachelor of Arts — Computer Science & Systems | Minor in Mathematics

December 2025

Clover Park Technical College
Associate in Applied Technology — Computer Programming

March 2022

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, SQL

Frontend: React, Next.js, Material-UI, Bootstrap

Backend: Node.js, Express, ASP.NET Core MVC, REST APIs, Microservices, JWT/Auth

Databases: PostgreSQL, SQLite, MySQL, Microsoft SQL Server

Testing: Unit Testing, BVT, Regression Testing, Automated Testing, Postman

Cloud & DevOps: AWS, Heroku, Render, Vercel

Other: Swagger/OpenAPI, Search Algorithms, Bayesian Inference

EXPERIENCE

Amazon Web Services | *Cloud Support Associate Intern*
Seattle, WA

June 2025 – September 2025

- Built a serverless file classification pipeline using S3, Lambda, CloudWatch, and IAM — automatically routed uploaded files to designated S3 buckets by file type via S3 event triggers, with CloudWatch logging for execution monitoring and IAM roles scoped to least-privilege access.
- Diagnosed and reproduced customer-reported failures across EC2, S3, and RDS, accelerating resolution workflows for the support engineering team.

Pioneer Technologies Corporation | *Software Tester & Developer*
Lacey, WA

February 2022 – February 2024

- Engineered and maintained a high-volume SQL database tracking 100,000+ records for a government-mandated clean water remediation program, performing daily delta processing and writing queries to ensure data integrity across dependent systems.
- Identified and resolved critical duplicate record issues alongside 2 coworkers, eliminating improperly housed resident entries and recovering \$1,000,000 in annual operational costs for the parent company.
- Developed and executed Python-based BVT and unit test suites for a customer-facing web portal, surfacing 30+ defects within the first month and establishing a repeatable daily regression baseline run every morning before business hours.
- Led QA operations for a team of 7 across the quarterly government-mandated letter distribution process, validating and deploying 300,000+ resident notifications while enforcing compliance standards for a North Carolina water contamination remediation effort.

PERSONAL PROJECTS

Full-Stack Movie & TV Show Application

Sept 2025 – Dec 2025

- Redesigned the PostgreSQL schema and SQL import pipeline, cutting database load time from 13 hours to under 3 minutes — a 99.6% performance improvement.
- Architected a RESTful API using Node.js, Express, and TypeScript with full CRUD support, deployed to Render with auto-deploy on push.
- Implemented a separate credentials microservice with JWT auth, RBAC, and email/SMS verification, deployed independently to Heroku.
- Developed search, pagination, and watchlist features on a React + Material-UI frontend deployed to Vercel, integrating across 3 independently hosted third-party APIs within a 4-person distributed team.
- Produced comprehensive Swagger/OpenAPI documentation covering all implemented routes for endpoint testing and team collaboration.

Program File Watcher

Jan 2025 – Mar 2025

- Designed a multi-threaded monitoring architecture using Java NIO WatchService with a dedicated daemon thread for blocking file events, keeping all Swing UI updates confined to the Event Dispatch Thread to prevent UI freezes and ensure thread safety.
- Built an advanced query interface with 6 filter modes (file name, path, extension, event type, date range, checkbox selection), predefined aggregate queries, and CSV export — backed by parameterized SQL to prevent injection.
- Implemented recursive directory registration using the Visitor pattern via SimpleFileVisitor to automatically register nested subdirectories at watch-start, with graceful exception handling to avoid crashes on restricted system paths.

Treasure Hunt AI Agent

Oct 2025 – Dec 2025

- Implemented and benchmarked BFS, DFS, UCS, A*, and Greedy Best-First Search across grid environments scaling from 10×10 to 30×30, comparing path cost, nodes expanded, and runtime across configurations.
- Built a Minimax adversarial agent with Alpha-Beta pruning and configurable depth control, reducing node expansion versus baseline Minimax while maintaining decision quality in competitive treasure collection scenarios.
- Engineered a Bayesian inference module with a noisy sensor model, maintaining and updating a full belief distribution over grid cells using Bayes' rule — tested across low, medium, and high noise levels averaged over 10 random seeds.
- Integrated all four AI modules into a unified decision layer that dynamically selects behavior based on environment state, opponent position, and belief confidence.

INTERESTS & ACHIEVEMENTS

- AWS Certified AI Practitioner (July 2025 – July 2028)
- Dean's List: UW Tacoma — 7 Quarters | Clover Park Technical College — 6 Quarters
- Academic Honor Society — Epsilon Pi Epsilon Member, UW Tacoma Chapter